

Redis Notes (How it works)

What is Redis?

Redis = Remote Dictionary Server

It is an **open-source, in-memory** data structure store.

Think of Redis as a **super-fast HashMap that lives in RAM instead of Disk.**

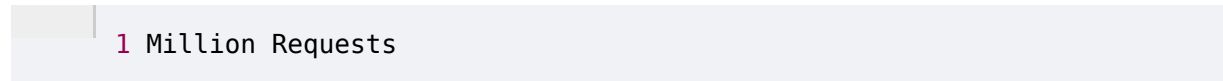
It can work as

- Cache
 - Database
 - Message Broker
 - Queue
 - Pub/Sub system
 - Session Store
 - Rate Limiter
 - Leaderboard
 - Distributed Lock
-

Why was Redis created?

Imagine this.

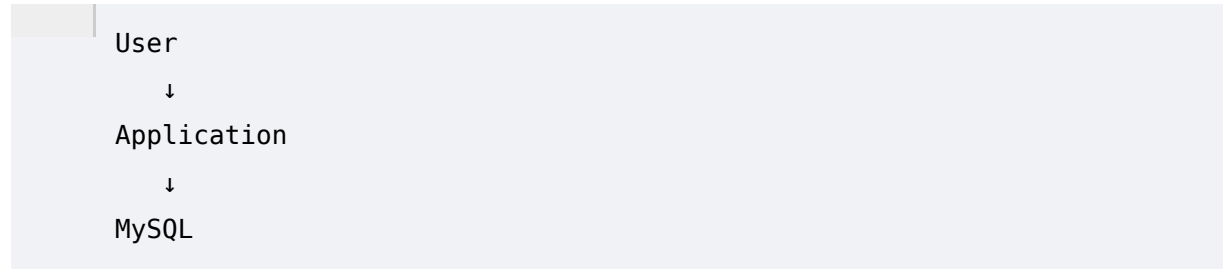
Your application receives



1 Million Requests

A light blue rectangular box with a thin vertical line on the left side. The text "1 Million Requests" is centered within the box.

Every request goes to MySQL.



```
graph TD; User --> Application; Application --> MySQL;
```

User
↓
Application
↓
MySQL

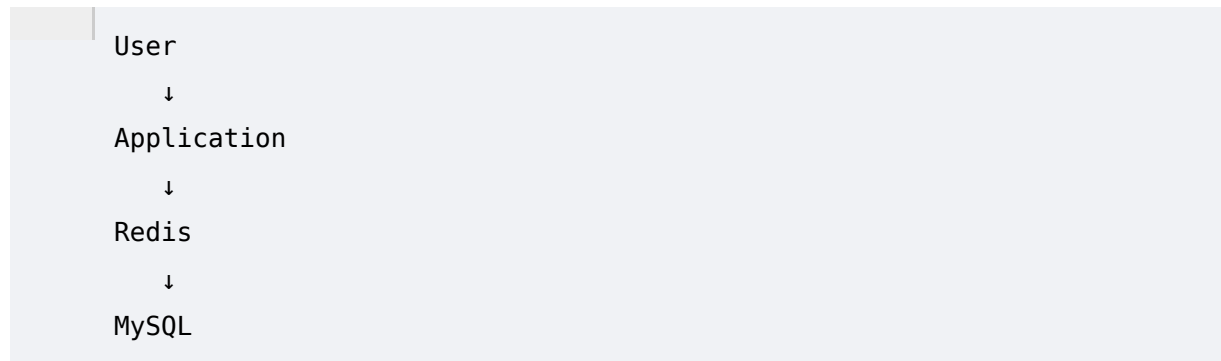
A light blue rectangular box containing a flow diagram. The diagram shows 'User' at the top, followed by a downward arrow to 'Application', followed by another downward arrow to 'MySQL'.

Problems

- Disk is slow
- Database becomes bottleneck
- More CPU usage

- High latency

Instead



Most requests are answered by Redis.

MySQL is touched only when necessary.

Result

- Very fast
- Low latency
- High throughput

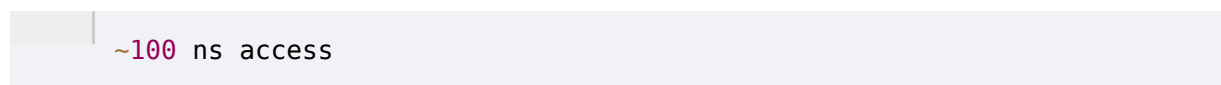
Why is Redis Fast?

This is one of the most asked interview questions.

Reason 1

Everything stays inside RAM.

RAM



SSD



Hard Disk



RAM is hundreds to thousands of times faster.

Reason 2

Single Threaded

Many people think

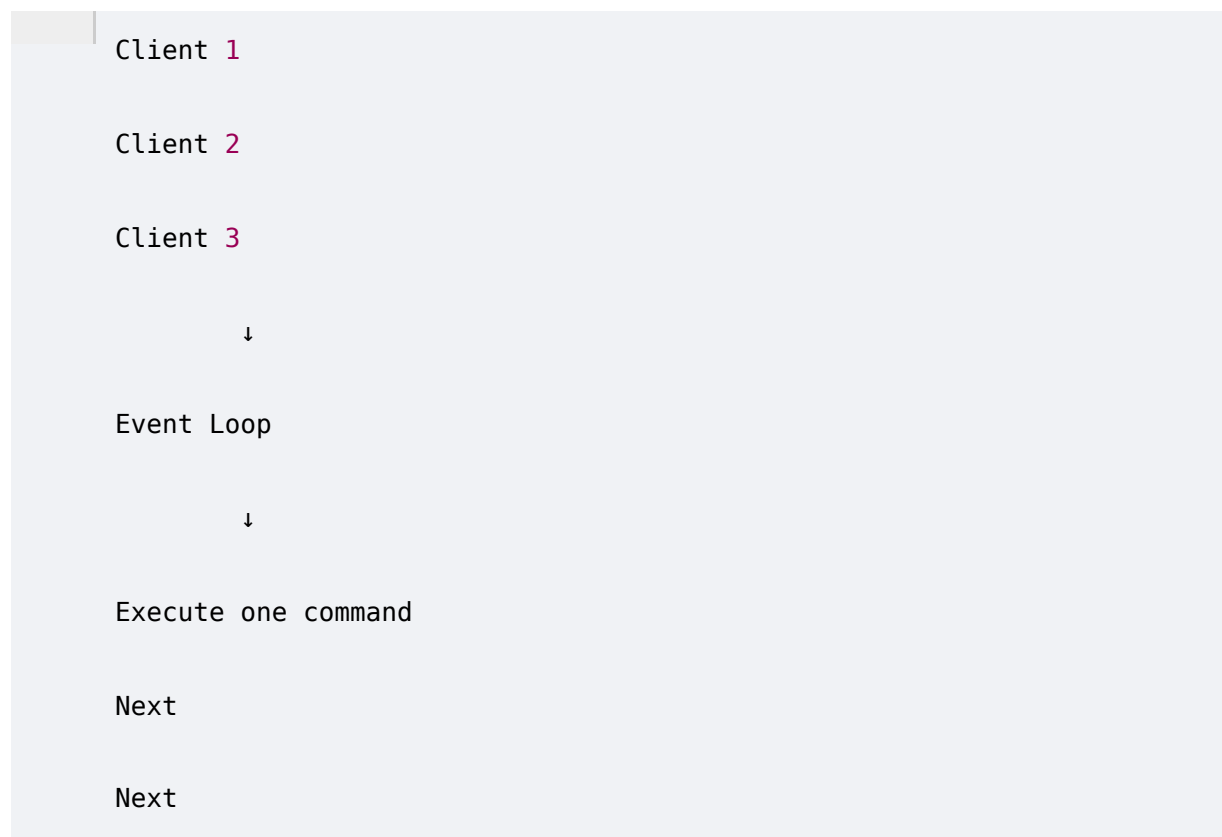
Single Thread = Slow

Wrong.

Redis avoids

- locks
- mutexes
- race conditions
- context switching

Instead

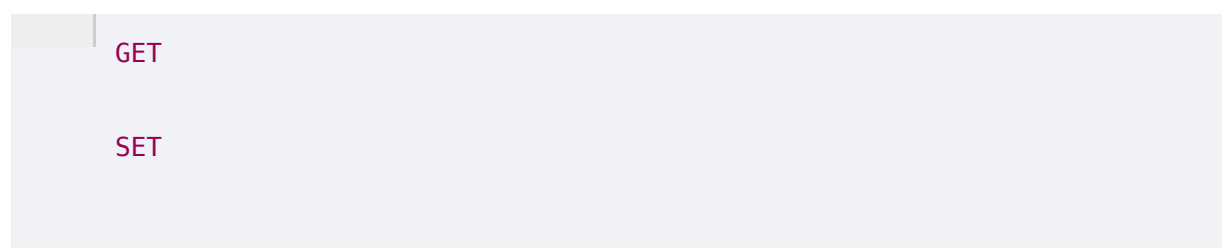


Very little overhead.

Reason 3

Very Simple Operations

Redis operations are mostly



DEL

INCR

LPUSH

HSET

Most of them are

$O(1)$

Time Complexity.

Reason 4

Efficient Memory Management

Redis uses optimized internal structures.

It stores data very compactly.

Less memory usage.

More speed.

Redis Architecture

Client

↓

Redis Server

↓

Memory

↓

(Optional)

Disk

Clients never access memory directly.

Everything goes through Redis.

Redis Request Flow

Step 1

Client sends

```
GET user:10
```

↓

Step 2

Redis receives request.

↓

Step 3

Redis looks inside RAM.

↓

Step 4

Value found.

↓

Step 5

Returns response.

```
{  
  name:"Alice"  
}
```

Entire process usually takes

```
Less than 1 millisecond
```

Redis is Single Threaded

Interview Question

Why doesn't Redis use multiple threads?

Answer

Because

Most work is

```
Memory Access
```

not CPU computation.

Using many threads would create

- Locking
- Synchronization
- Context Switching

which actually slows things down.

Does Redis Really Use One Thread?

Modern Redis does use background threads for

- Persistence
- Lazy deletion
- I/O optimization

BUT

Command execution

```
SET
```

```
GET
```

```
DEL
```

is still single-threaded.

Redis Data Types

Redis is NOT just Key-Value.

It supports many structures.

1. String

Most common.

```
SET name Chandan
```

GET name

DEL name

Can store

- Text
- Number
- JSON
- Binary

Examples

User Name

OTP

JWT Token

Access Token

Cache

2. Hash

Like

unordered_map

or

Dictionary

Example

User

Name

Age

City

Commands

HSET

HGET

HDEL

Example

user:1

name

Alice

age

25

Instead of many keys

user1_name

user1_age

user1_city

Use one hash.

3. List

Ordered collection.

Supports

LPUSH

RPUSH

LPOP

RPOP

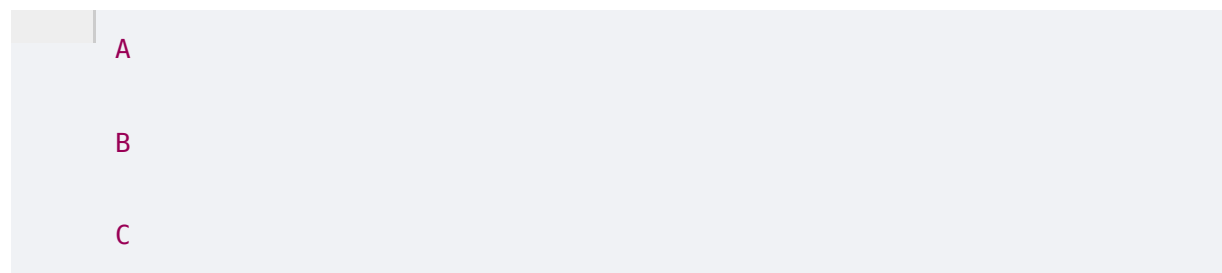
Uses

- Queue
 - Stack
 - Recent Searches
 - Notifications
-

4. Set

Unique values.

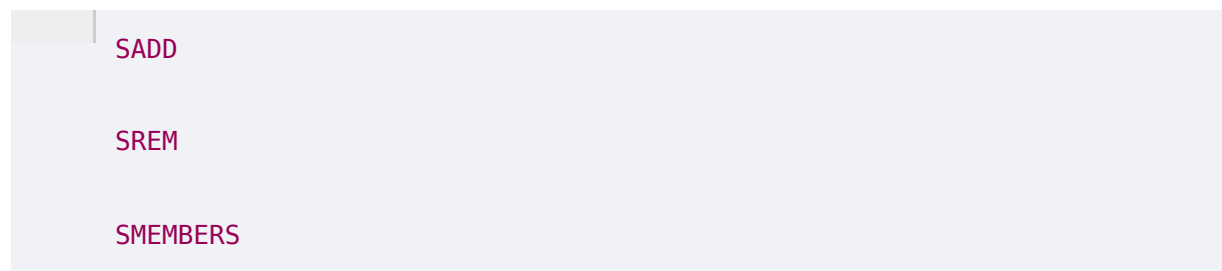
No duplicates.



Useful for

- Friends
- Followers
- Tags
- Unique Visitors

Commands



5. Sorted Set

Like Set

BUT

Every item has



Example

```
Alice 90
```

```
Bob 70
```

```
John 95
```

Automatically sorted.

Used in

- Leaderboards
- Rankings
- Top Players

Commands

```
ZADD
```

```
ZRANGE
```

```
ZREM
```

6. Stream

Like Kafka (small version).

Useful for

- Event Processing
- Logs
- Messaging

7. Bitmap

Stores bits.

```
001001101
```

Used for

- Attendance
- Daily Login

- Feature Flags

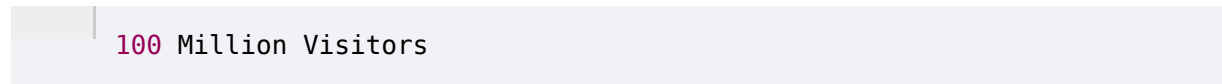
Very memory efficient.

8. HyperLogLog

Very famous interview topic.

Problem

Need to count



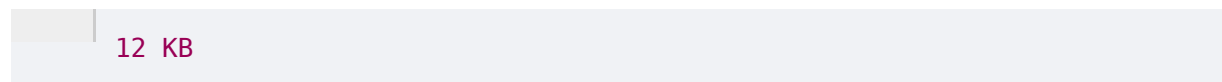
Using Set

Huge memory.

HyperLogLog

Uses

Around



Memory.

Approximate answer.

Used in

Google Analytics style counting.

9. GeoSpatial

Stores

Latitude

Longitude

Find nearby

- Restaurants
 - Drivers
 - Hospitals
-

Persistence

Redis stores data in RAM.

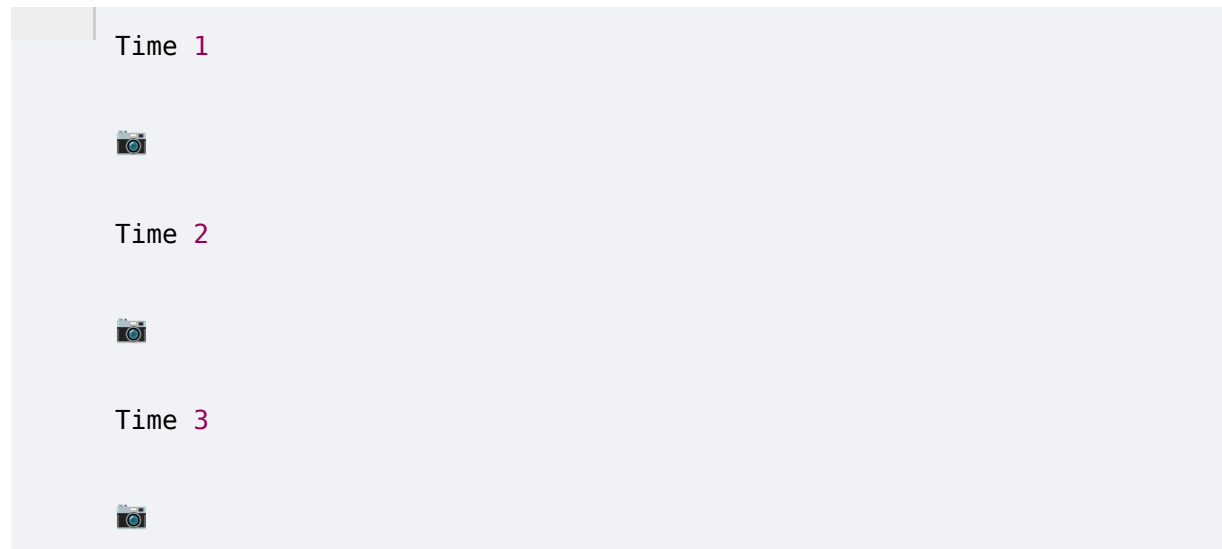
What if server crashes?

Need persistence.

Redis has two options.

RDB Snapshot

Think of taking photos.



Advantages

- Small file
- Faster startup
- Backup friendly

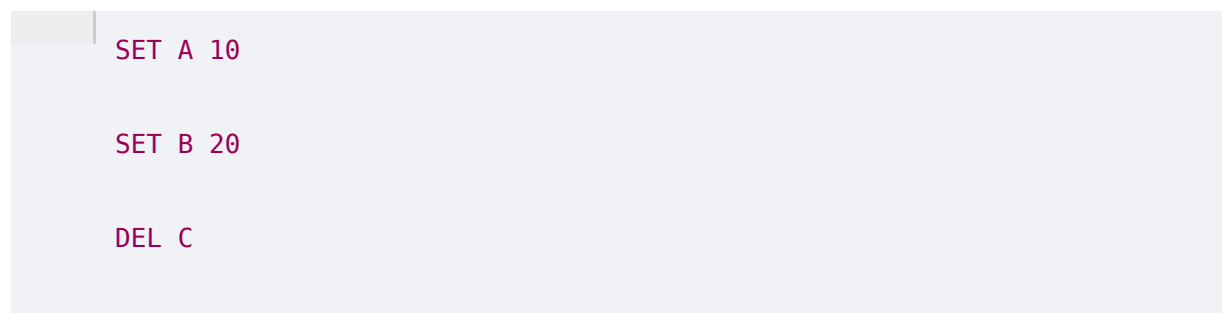
Disadvantage

Lose recent data.

AOF (Append Only File)

Every write command is written.

Example



INCR D

Restart?

Replay commands.

Advantages

Very little data loss.

Disadvantage

Larger files.

Which is Better?

RDB

Good for

Backups

AOF

Good for

Durability

Production usually uses

Both

Cache

Most important Redis use case.

Without Redis

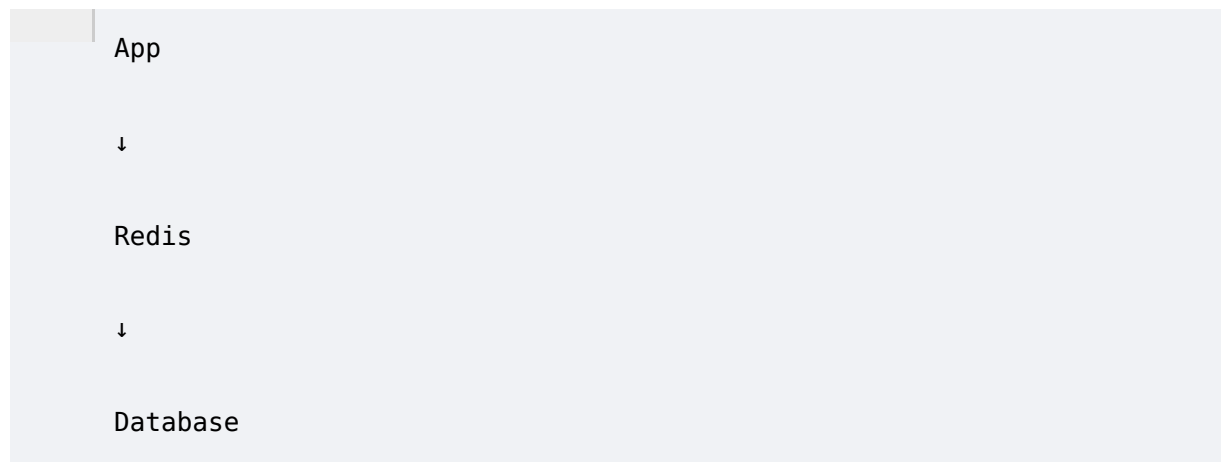
App

↓

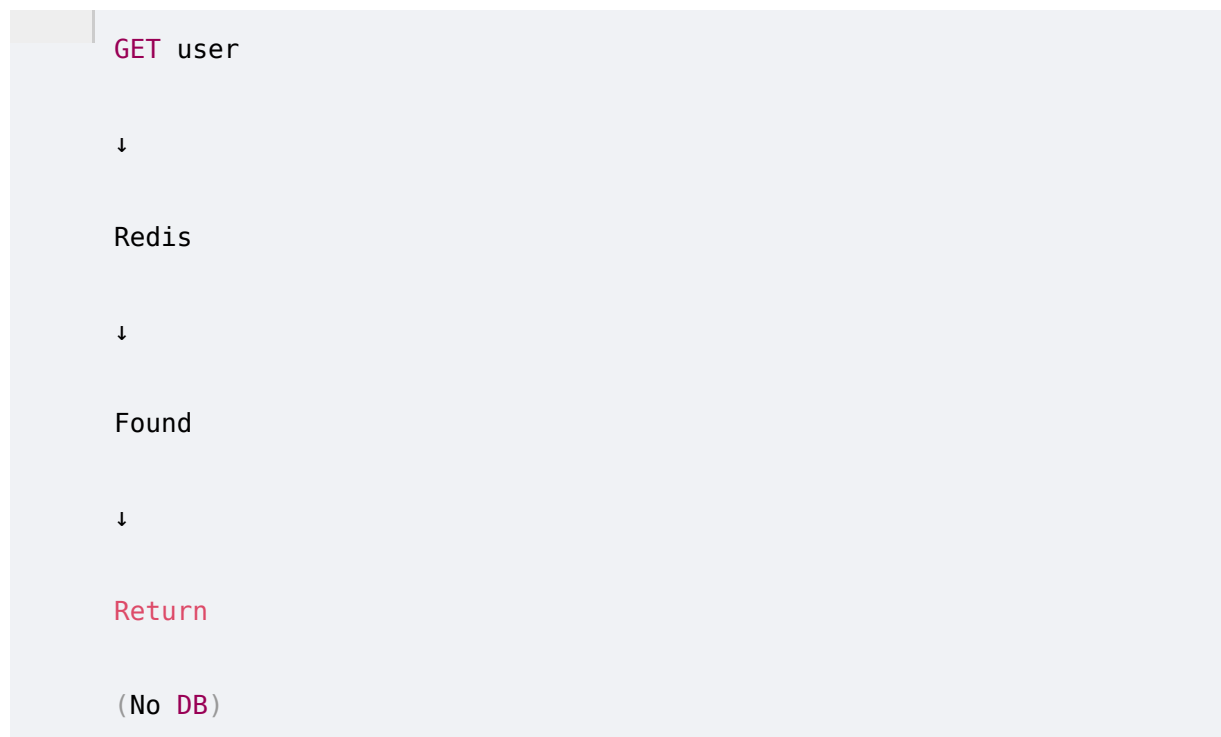
Database

Every request.

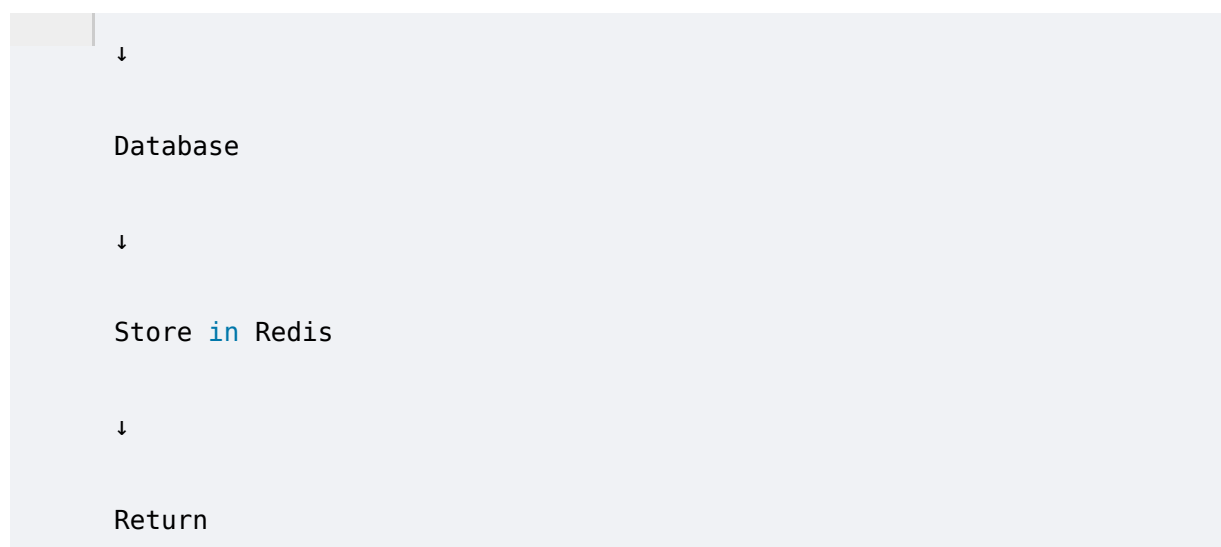
With Redis



Flow



If not found



Called

```
Cache Aside Pattern
```

Very common interview topic.

Cache Hit

```
Data found in Redis
```

Fast.

Cache Miss

```
Data absent
```

```
↓
```

```
Read DB
```

```
↓
```

```
Store in Redis
```

TTL

Time To Live.

Example

```
OTP
```

```
5 minutes
```

Command

```
EXPIRE otp 300
```

After

300 seconds

Automatically deleted.

Eviction Policy

Redis memory becomes full.

What should Redis remove?

Policies

LRU

Least Recently Used

Remove least used data.

Most popular.

LFU

Least Frequently Used

Remove rarely accessed data.

TTL

Remove keys that expire first.

Pub/Sub

Publisher

↓

Redis

↓

Subscribers

Example

Chat

Notification

Live Score

Publisher doesn't know subscribers.

Very loosely coupled.

Rate Limiting

Suppose

User can call API

```
100 requests
```

```
per minute
```

Redis

```
INCR
```

```
EXPIRE
```

Easy implementation.

Session Storage

Instead of storing sessions

inside application

Store

```
Session
```

```
↓
```

```
Redis
```

Benefits

- Faster
 - Shared across servers
 - Stateless application
-

Leaderboard

```
Player
```

```
↓
```

```
Sorted Set
```

↓

Rank

Operations

Top 10

Top 100

Player Rank

All extremely fast.

Distributed Lock

Suppose

Two servers

Server A

Server B

Both try to update same order.

Redis

```
SET lock NX EX 30
```

Only one gets lock.

Very common interview question.

Replication

Master

↓

Replica

↓

Replica

Advantages

- High Availability
 - Read Scaling
 - Backup
-

Redis Cluster

Single Redis

↓

Memory Full

Problem.

Cluster

Redis 1

Redis 2

Redis 3

Redis 4

Data split across nodes.

Benefits

- Horizontal Scaling
 - Fault Tolerance
-

Common Commands

SET key value

GET key

DEL key

EXISTS key

EXPIRE key 60

TTL key

INCR counter

DECR counter

HSET

Time Complexities (Very Important)

Operation	Complexity
GET	$O(1)$
SET	$O(1)$
DEL	$O(1)$
HGET	$O(1)$
HSET	$O(1)$
LPUSH	$O(1)$
RPUSH	$O(1)$
SADD	$O(1)$
ZADD	$O(\log N)$
ZRANGE	$O(\log N + M)$

Advantages

- ✓ Extremely fast
- ✓ Low latency
- ✓ In-memory
- ✓ Rich data structures

- ✓ Easy to use
 - ✓ Great cache
 - ✓ Supports persistence
 - ✓ Replication
 - ✓ Clustering
 - ✓ Pub/Sub
-

Disadvantages

- ✗ RAM is expensive.
 - ✗ Data may be lost without persistence.
 - ✗ Single-threaded command execution can become CPU-bound for very heavy workloads.
 - ✗ Not suitable for complex relational queries.
 - ✗ Not a replacement for SQL databases.
-

FAANG Interview Questions

Beginner (0–2 Years)

1. What is Redis?
 2. Why is Redis fast?
 3. Why does Redis use RAM?
 4. Difference between Redis and MySQL.
 5. What is caching?
 6. Cache Hit vs Cache Miss.
 7. What is TTL?
 8. Explain Redis architecture.
 9. Explain String, Hash, List, Set, and Sorted Set.
 10. Why is Redis called a data structure store?
-

Intermediate (2–5 Years)

1. Why is Redis single-threaded?
2. RDB vs AOF.
3. Redis Replication.
4. Redis Cluster.

5. LRU vs LFU eviction.
6. How does Pub/Sub work?
7. How would you implement rate limiting using Redis?
8. How would you build a leaderboard?
9. How do distributed locks work using Redis?
10. What happens when Redis memory becomes full?